

Section	Purpose
Name	Shortage order issue
Description	Due to item shortage (the inventory reaches its min quantity) the system sends a notification with a report to make a new order depending on the info that has been received
Actors	Suppliers, Suppliers Manager
Precondition	The system received a notification about an inventory shortage, and both the supplier and the product are already registered in the system.
Post-conditions	A new order was created due to the shortage, and the supplier received the order.
Basic course of action	1. The system receives a shortage report, including product details and the requested quantity of it. 2. The system makes a new order according to the report. 3. The system searches for the cheapest supplier for the order using the bill of quantities. 4. If there are more than one supplier with the same cheapest price, the system chooses a supplier randomly.
Alternate courses	1. More than one supplier with the cheapest price. 2.1. The product is not available

Pseudocode:

```

if ShortageExist :
    report <- getShortageReport(items)
    Order NewOrder <- placeOrder(reportItems)
    SuppList <- getSystemSuppliersDependingOnOrder(NewOrder)
    List cheapestSupp <- getCheapestSupplier(SuppList, NewOrder)
    if len cheapestList == 0:
        throw exception "-No supplier found"
    if len cheapestSupp == 1:
        Supplier newSupp <- cheapestSupp[0]
        MakeOrder(newSupp, newOrder)
    // if there are more than one supplier choose one randomly and then do the same

```